

CAD Design Report – Automatic Curtain Opener

1. Introduction

The CAD (Computer-Aided Design) stage is a vital part of the design and development process of the Automatic Curtain Opener. It allows for the digital prototyping of the mechanical system, enabling the visualization, modification, and testing of components before physical implementation. In this project, CAD tools were used to design the structural framework and moving elements of the curtain opener, ensuring that the system was both functional and compact. The CAD model serves as a blueprint for fabrication and assembly, as well as a foundation for simulation and 3D printing. While doing the design of the project i was also able to study about many feature and tools that are used in the application which is really user friendly and helps the user to design a model with ease. The app also Fusion 360 also allows the user to design the work in more accurate manner rather than a simple design.

2. Objectives

The primary objectives of the CAD portion of this project were:

- To create accurate 3D models of all mechanical components of the curtain opener.

- To ensure that all parts fit together with proper tolerances and alignment.
- To simulate the movement of the curtain and verify that the design operates smoothly without collision.
- To minimize the overall footprint of the mechanism for aesthetic integration in home interiors.
- To ensure a smooth movement and working of the curtains

3. Tools and Software

The CAD models were developed using the following tools:

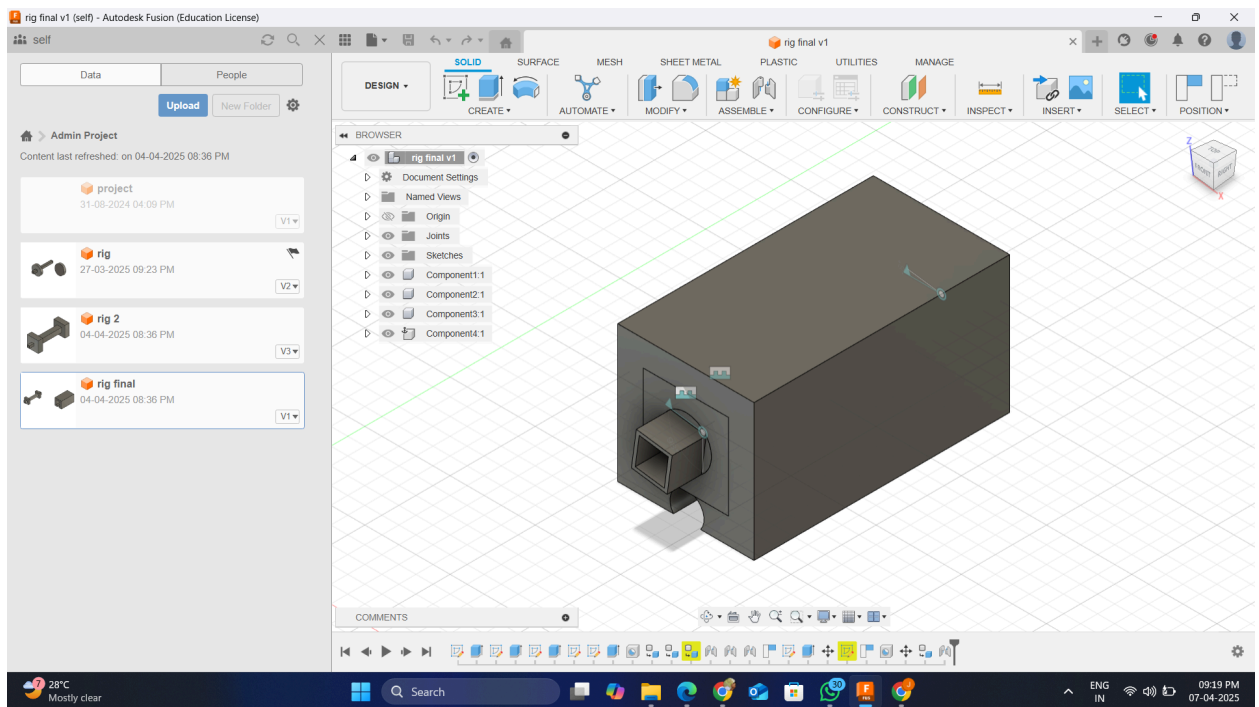
- **Software Used:** Autodesk Fusion 360
- **Modeling Approach:** Parametric modeling to allow easy adjustment of dimensions.
- **Simulation Features Used:** Motion simulation
- **Export Formats:** `.f3d` for internal editing, `.stl` for 3D printing, and `.step` for hardware sharing.

4. Components Designed and step by step procedure

1. **Simple vertical rail :**At first I designed a simple vertical rail that would fit into the 12 V dc motor perfectly.Since the rail is vertical i first sketched the basic outline for the design.The design included a box like portion which would correctly fit to the dc motor and a

cylindrical portion attached to it so when the motor moves the cylindrical rail also moves along with it.

- 2. Joints :** I added a simple revolute joint for the cylindrical part which was inside two rectangular shaped boxes. The motion of the joint can be viewed by animating the joint. I also added a rigid joint between the two rectangular portion and the outer casing.
- 3. Casing:** The upper limit switch is fixed to the casing and the casing also ensures smooth movement of the cloth in the curtain. Also I Have added a cylindrical shaped hole in the casing that would fit the end of the curtain perfectly as nowadays in vertical curtains there is a cylindrical part at the end. The main reason for the design of casing is to fix the upper limit switch in a suitable position so that the cylindrical portion correctly closes the limit switch and turns off the curtain.



5. Design Considerations

During the modeling process, several key factors were taken into account:

Precision: The tolerances between moving parts and mounts were adjusted to accommodate 3D printing inaccuracies.

Modularity: The design supports quick assembly and disassembly for maintenance and upgrades.

Compactness: Emphasis was placed on keeping the design unobtrusive for integration into domestic interiors.

7. Output and Fabrication

- All custom parts were exported as `.stl` files for 3D printing.

Things I learned from the overall project and my experience in RIG lab

1. **Electronics part:** In the electronics part I learned how to use NodeMCU (ESP8266) where it acts as the brain of the entire project we could code our requirements into the NodeMCU (ESP8266) and it works accordingly. The software used for coding

into the NodeMCU was arduino. I also gained practical knowledge about integrating various components such as the L298N motor driver, DC motor, and limit switches. I learned how to control motor movement using digital signals and how to set up a web server on the ESP32 for Wi-Fi-based control and make the NodeMCU as hotspot setup. Understanding the role of each component in the circuit and how they interact helped me develop troubleshooting skills and improve my wiring techniques. This part of the project also deepened my understanding of automation, circuit design, power management, and real-time system response.

2. **Programming part:** Working with MIT App Inventor as part of the Automatic Curtain Opener project helped me understand how to create interactive mobile applications without needing advanced programming skills. I learned how to design a user-friendly interface with buttons, labels, and input fields to control the curtain system wirelessly over Wi-Fi. I also learned about the use of Web components to send HTTP requests to the ESP32, which allowed the mobile app to communicate with the hardware. The step by step code blocks made it easier to grasp event-driven programming concepts, such as triggering actions when buttons are clicked or when data is received. I also learned how to handle responses from the ESP32 and update the status on the app in real-time. This experience improved my skills in mobile app design, logical thinking, and connectivity between hardware and software. Overall, it gave me the confidence to build more robotic applications.

MY EXPERIENCE IN RIG LAB

My project was to build an automatic curtain opener and my teammates were Anton and Srimukha. We started the work of the project from the

day after our first meeting my teammates were very friendly and cooperative and we enjoyed the time we spent in RIG lab. I have to specially mention the seniors and our mentors who have helped us complete this project. The seniors of the club were actively promoting the juniors to do their projects and was helping us to learn new things in the world of robotics. When our project was on the way to completion we could feel the happiness. At last when our project was completed we couldn't even believe that it was possible for us. I personally was really happy when the project was completed and was working perfectly according to what we have planned. In overall it was a different experience for me in the lab learning new things and enjoying it along the way.

8. Conclusion

The CAD design phase was instrumental in ensuring the success of the Automatic Curtain Opener project. It enabled precise planning, simulation, and production of essential mechanical components. Through digital modeling, potential issues were identified and resolved before physical assembly, saving time and material. The modular and customizable nature of the CAD design allows for future upgrades, including larger motors, different curtain lengths, or enhanced enclosure designs. Overall, the CAD work provided a strong foundation for the integration of electronics, software, and mechanical motion in this smart home automation system.

